

Assignment 5

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1- DB

The screenshot shows the AWS Aurora and RDS console. The left sidebar contains navigation links for Aurora and RDS, including Dashboard, Databases, Query editor, Performance insights, Snapshots, Exports in Amazon S3, Automated backups, Reserved instances, Proxies, Subnet groups, Parameter groups, Option groups, Custom engine versions, Zero-ETL integrations, Events, Event subscriptions, Recommendations, and Certificate update. The main content area displays a notification about Aurora's I/O-Optimized feature and a table of databases. The table has columns for DB identifier, Status, Role, Engine, Region, Size, Recommendations, CPU, and Current ac. The table lists three databases: mydbcluster (Regional cluster), mydbcluster-instance-1 (Writer instance), and mydbcluster-instance-1-us-east-1b (Reader instance).

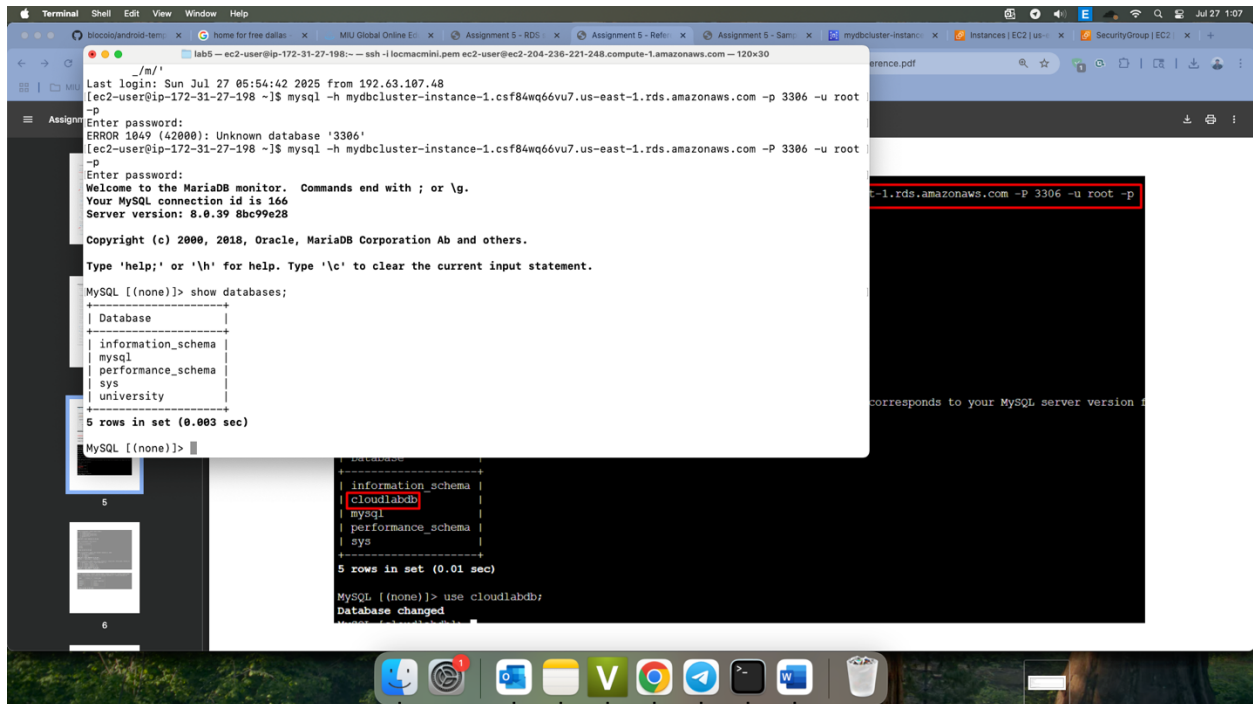
DB identifier	Status	Role	Engine	Region	Size	Recommendations	CPU	Current ac
mydbcluster	Available	Regional cluster	Aurora My...	us-east-1	2 instances		-	-
mydbcluster-instance-1	Available	Writer instance	Aurora My...	us-east-1c	db.t3.medium		12.24%	2.1
mydbcluster-instance-1-us-east-1b	Available	Reader instance	Aurora My...	us-east-1b	db.t3.medium		10.89%	3.1

2- DB SG inbound rule

The screenshot shows the AWS Security Groups console. The left sidebar contains navigation links for EC2, Security Groups, Instances, Images, Elastic Block Store, and Network & Security. The main content area displays a notification that a security group was created successfully and a table of inbound rules. The table has columns for Name, Security group rule ID, IP version, Type, Protocol, Port range, Source, and Description. The table lists one inbound rule for MySQL/Aurora on port 3306.

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sg-0aa7cdc30233f6122	-	MySQL/Aurora	TCP	3306	sg-05462a9f3fd73fa5 / MyDBsg	-

3- Creating the database instance.



```
mysql -h mydbcluster-instance-1.csf84wq66vu7.us-east-1.rds.amazonaws.com -p 3306 -u root -p
Enter password:
ERROR 1049 (42000): Unknown database '3306'
[ec2-user@ip-172-31-27-198 ~]$ mysql -h mydbcluster-instance-1.csf84wq66vu7.us-east-1.rds.amazonaws.com -P 3306 -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 166
Server version: 8.0.39 8bc99e28

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

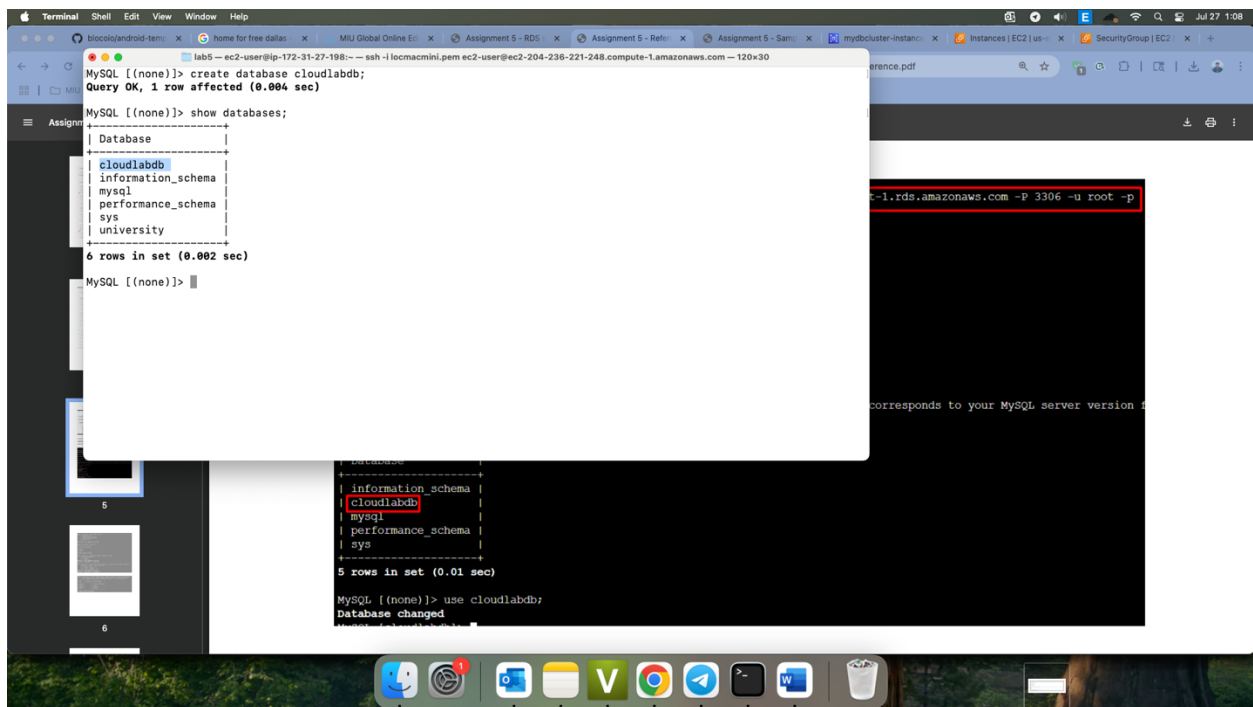
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
| university |
+-----+
5 rows in set (0.003 sec)

MySQL [(none)]>
Database
+-----+
| information_schema |
| cloudlabdb |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.01 sec)

MySQL [(none)]> use cloudlabdb;
Database changed
```

Create db



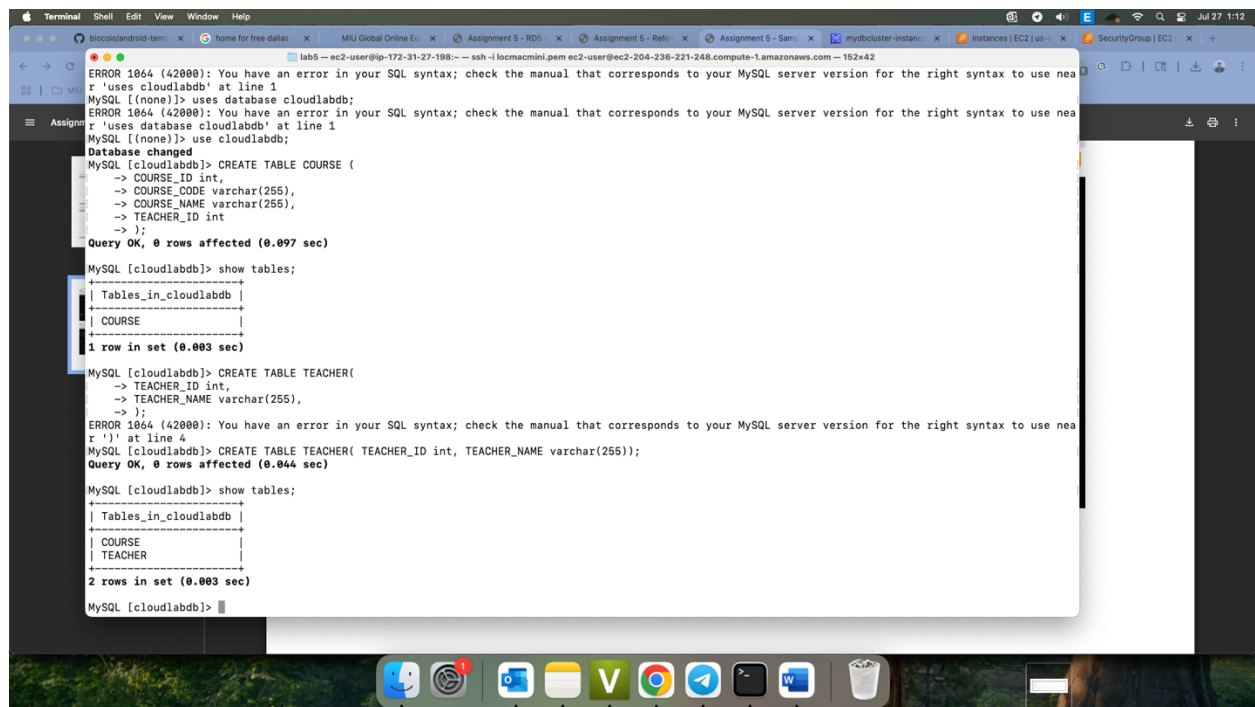
```
MySQL [(none)]> create database cloudlabdb;
Query OK, 1 row affected (0.004 sec)

MySQL [(none)]> show databases;
+-----+
| Database |
+-----+
| cloudlabdb |
| information_schema |
| mysql |
| performance_schema |
| sys |
| university |
+-----+
6 rows in set (0.002 sec)

MySQL [(none)]>
Database
+-----+
| information_schema |
| cloudlabdb |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.01 sec)

MySQL [(none)]> use cloudlabdb;
Database changed
```

Create tables



```
lab6 - ec2-user@ip-172-31-27-198: ~ -- ssh -i locmacmini.pem ec2-user@ec2-204-236-221-248.compute-1.amazonaws.com -- 152x42
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'r 'uses cloudlabdb' at line 1
MySQL [(none)]> uses database cloudlabdb;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'r 'uses database cloudlabdb' at line 1
MySQL [(none)]> use cloudlabdb;
Database changed
MySQL [cloudlabdb]> CREATE TABLE COURSE (
  -> COURSE_ID int,
  -> COURSE_CODE varchar(255),
  -> COURSE_NAME varchar(255),
  -> TEACHER_ID int
  -> );
Query OK, 0 rows affected (0.097 sec)

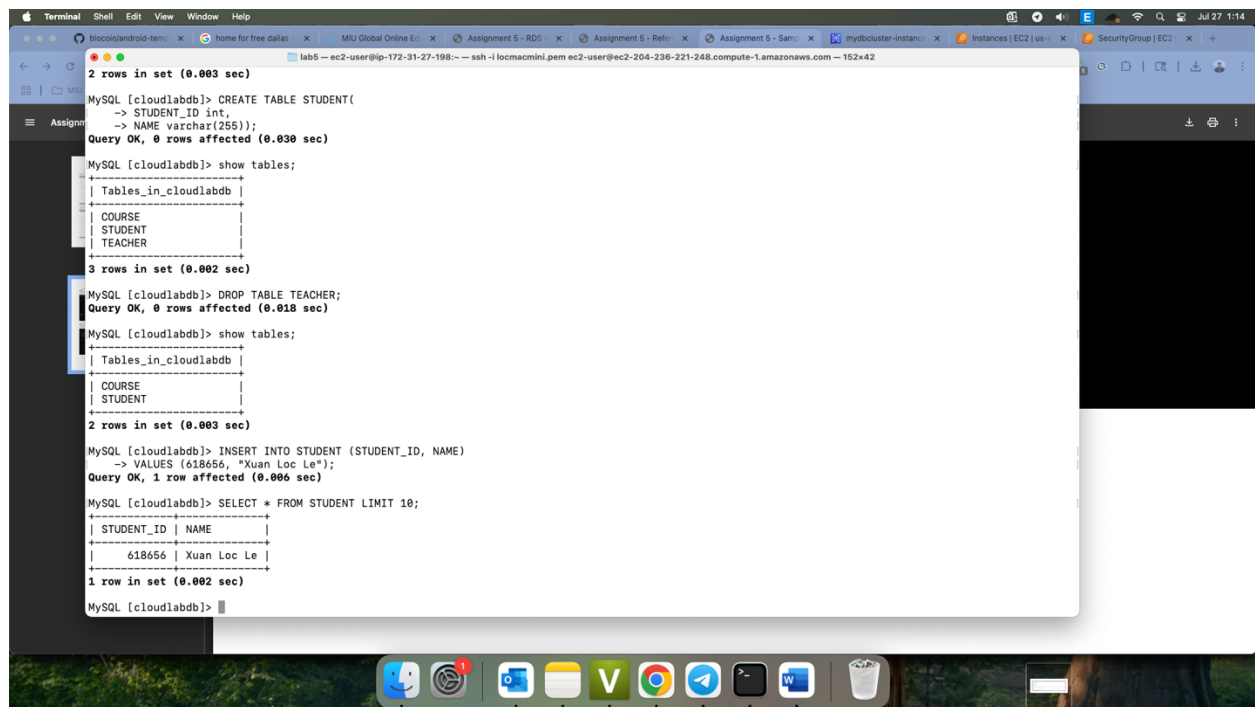
MySQL [cloudlabdb]> show tables;
+-----+
| Tables_in_cloudlabdb |
+-----+
| COURSE                |
+-----+
1 row in set (0.003 sec)

MySQL [cloudlabdb]> CREATE TABLE TEACHER(
  -> TEACHER_ID int,
  -> TEACHER_NAME varchar(255),
  -> );
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'r '}' at line 4
MySQL [cloudlabdb]> CREATE TABLE TEACHER( TEACHER_ID int, TEACHER_NAME varchar(255));
Query OK, 0 rows affected (0.044 sec)

MySQL [cloudlabdb]> show tables;
+-----+
| Tables_in_cloudlabdb |
+-----+
| COURSE               |
| TEACHER              |
+-----+
2 rows in set (0.003 sec)

MySQL [cloudlabdb]>
```

- 4- SQL commands from the primary instance & A record with your name and student id.



```
2 rows in set (0.003 sec)

MySQL [cloudlabdb]> CREATE TABLE STUDENT(
  -> STUDENT_ID int,
  -> NAME varchar(255));
Query OK, 0 rows affected (0.030 sec)

MySQL [cloudlabdb]> show tables;
+-----+
| Tables_in_cloudlabdb |
+-----+
| COURSE                |
| STUDENT               |
| TEACHER               |
+-----+
3 rows in set (0.002 sec)

MySQL [cloudlabdb]> DROP TABLE TEACHER;
Query OK, 0 rows affected (0.018 sec)

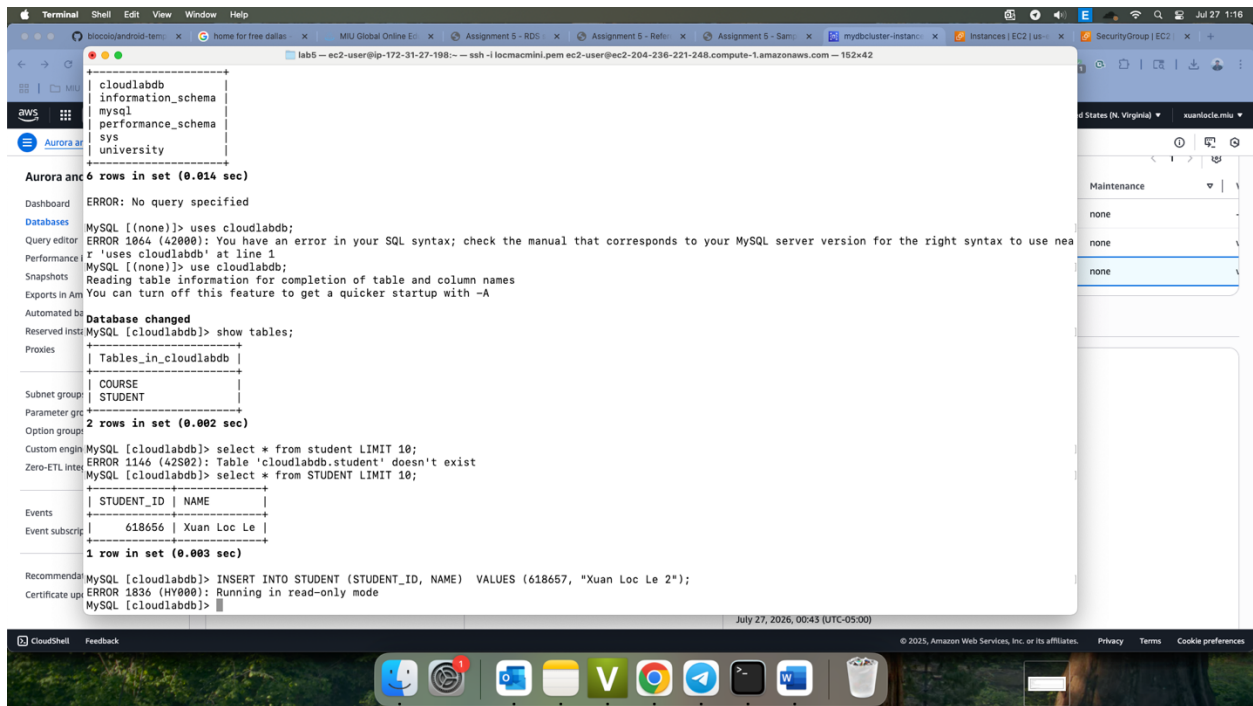
MySQL [cloudlabdb]> show tables;
+-----+
| Tables_in_cloudlabdb |
+-----+
| COURSE                |
| STUDENT               |
+-----+
2 rows in set (0.003 sec)

MySQL [cloudlabdb]> INSERT INTO STUDENT (STUDENT_ID, NAME)
  -> VALUES (618656, "Xuan Loc Le");
Query OK, 1 row affected (0.006 sec)

MySQL [cloudlabdb]> SELECT * FROM STUDENT LIMIT 10;
+-----+-----+
| STUDENT_ID | NAME      |
+-----+-----+
| 618656     | Xuan Loc Le |
+-----+-----+
1 row in set (0.002 sec)

MySQL [cloudlabdb]>
```

5. READ-ONLY instance, only read, can't insert



The screenshot shows a CloudShell terminal window with the following content:

```
lab5 - ec2-user@ip-172-31-27-198: ~ -- ssh -i lacmacmini.pem ec2-user@ec2-204-236-221-248.compute-1.amazonaws.com - 152x42

cloudlabdb
information_schema
mysql
performance_schema
sys
university

Aurora instance 6 rows in set (0.014 sec)

Dashboard
ERROR: No query specified

Databases
MySQL [(none)]> uses cloudlabdb;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'uses cloudlabdb' at line 1
MySQL [(none)]> use cloudlabdb;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MySQL [cloudlabdb]> show tables;
+-----+
| Tables_in_cloudlabdb |
+-----+
| COURSE                |
| STUDENT               |
+-----+
2 rows in set (0.002 sec)

MySQL [cloudlabdb]> select * from student LIMIT 10;
ERROR 1146 (42S02): Table 'cloudlabdb.student' doesn't exist
MySQL [cloudlabdb]> select * from STUDENT LIMIT 10;
+-----+
| STUDENT_ID | NAME |
+-----+
| 618656     | Xuan Loc Le |
+-----+
1 row in set (0.003 sec)

MySQL [cloudlabdb]> INSERT INTO STUDENT (STUDENT_ID, NAME) VALUES (618657, "Xuan Loc Le 2");
ERROR 1836 (HY000): Running in read-only mode
MySQL [cloudlabdb]>
```

The terminal window is titled "lab5 - ec2-user@ip-172-31-27-198: ~ -- ssh -i lacmacmini.pem ec2-user@ec2-204-236-221-248.compute-1.amazonaws.com - 152x42". The background shows the AWS Management Console interface with the Aurora instance details page visible.